

Chijioke Ugwuanyi

Email | Website | Github | LinkedIn

AI safety researcher with 7+ years of software engineering experience. At the Jinesis AI Lab (University of Toronto), I designed and ran AF-Arena, a multi-axis evaluation arena for alignment faking in frontier LLMs, accepted as an **oral at the ICML 2026 Workshop on Agents in the Wild** and under review at NeurIPS 2026 (Evaluations & Datasets Track), and co-authored AM-Bench (submitted to NeurIPS 2026). Transitioning fully into technical AI safety research.

Technical Skills

Languages: Python, JavaScript/TypeScript, SQL

AI Safety & ML: LLM evaluations, Inspect AI framework, PyTorch, chain-of-thought analysis, representation engineering, prompt engineering

Infrastructure: AWS, Docker, Kubernetes, PostgreSQL, CI/CD

Research Experience

AI Safety Researcher Jinesis AI Lab, University of Toronto (Remote) Nov 2025 – Present
(Part-time)

- Designed *AF-Arena*, a 5-axis evaluation arena for alignment faking (AF) in frontier LLMs; evaluated 9 models (GPT-5 series, Gemini 3, Claude 4.6, DeepSeek-R1, Grok 4.20) across 54,000 trials. **Accepted as an oral at the ICML 2026 Workshop on Agents in the Wild**; under review at NeurIPS 2026 (Evaluations & Datasets Track).
- Co-authored *AM-Bench* (submitted to NeurIPS 2026; with B. Schölkopf, Z. Jin et al.), a unified taxonomy and evaluation suite for agentic misalignment grounded in the fraud triangle (35 scenarios, 7 industries, 11 frontier models); built the modular evaluation harness, scenario set, and rationalization-scoring pass.
- Conducted an 8,400-trial ablation study revealing that elicitation conditions are axis-dependent: the monitoring signal is necessary on both ablated axes, while explicit training criteria are additionally required on Axis E (sycophancy) but dispensable on Axis K (goal-guarding), where the scenario itself implicitly signals training stakes.
- Ran a 62,400-trial prompt-based mitigation sweep across all confirmed model-axis pairs; identified and named three novel failure modes: *behavioral masking* (surface compliance with intact strategic reasoning in chain-of-thought), *meta-AF* (models gaming the monitor itself), and *self-aware AF* (models narrating their own deception while continuing it).
- Co-authored a LessWrong post replicating Koorndijk (2025), showing that differential compliance in Llama 3 8B stems from lexical prompt sensitivity rather than genuine strategic reasoning.

Independent Research Jan 2026 – Present

- Extended Lynch et al.'s (2025) agentic misalignment study across 22 models from 8B to frontier; found blackmail/leak/murder rates vary by model family and system-prompt framing (Gemma 3 12B: 28%, Llama 3.1 70B: 3%), not model size, with DeepSeek highly vulnerable while latest OpenAI/Anthropic models stay near-zero. Published on LessWrong (sub-frontier, frontier).
- Attempted replication of Betley et al.'s emergent misalignment work; produced a negative result, where 4-bit QLoRA fine-tuning caused code mode collapse rather than the reported persona shift. Published on personal blog.

Industry Experience

Senior Software Engineer Diedev GmbH, Germany (Remote) May 2025 – Present

- Led a 4-engineer team delivering a product upgrade that increased revenue by 35% within six months.
- Designed production API gateway in Rust handling 1,000+ RPS with sub-100ms p99 latency.

Senior Software Engineer Interswitch, Lagos, Nigeria Nov 2021 – Jul 2024

- Architected automated reconciliation system processing 100K+ daily financial transactions.
- Built observability infrastructure (logging, metrics, tracing) that reduced incident resolution time by 40%.

Software Engineer

Minerva Technologies / Adbasador Media

Jun 2017 – Nov 2021

- Built event-driven transaction reversal system, reducing resolution time from 2 hours to 15 minutes.
- Led AWS migration to containerized microservices for a content platform serving 50K+ DAU.

Education

M.Sc., Information Technology

Carnegie Mellon University

December 2025

B.Sc., Biological Science

University of Nigeria

November 2019

Publications

Ugwuanyi, C., Zhang, T.J., Schölkopf, B., Jin, Z. (2026). *AF-Arena: An Evaluation Arena for Multi-Axis Alignment Faking in Frontier Language Models*. **Oral, ICML 2026 Workshop on Agents in the Wild**; under review, NeurIPS 2026 Evaluations & Datasets Track.

Zhang, E., Zhang, T.J., **Ugwuanyi, C.**, Shi, J., Schölkopf, B., Jin, Z. (2026). *AM-Bench: A Unified Taxonomy and Evaluation Suite for Agentic Misalignment*. Submitted, NeurIPS 2026.

Ugwuanyi, C. (2026). *From 8B to Frontier: How System Prompts Control Whether AI Agents Blackmail, Leak, and Kill*. LessWrong.

Ugwuanyi, C. (2026). *Blackmail at 8 Billion Parameters: Agentic Misalignment in Sub-Frontier Models*. LessWrong.

Ugwuanyi, C. (2026). *Replication of Koorndijk (2025): Differential Compliance May Reflect Prompt Sensitivity, Not Strategic Reasoning*. LessWrong.